

Acid–base indicators (HL)

Higher level (HL)

Learning outcomes

By the end of this section you should be able to:

- (R3.1.14) Construct equilibria expressions to show why the colour of an indicator changes with pH.
- (R3.1.15) Identify an appropriate indicator for a titration from the identity of the salt and the pH range of the indicator.

In section R3.1.8 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ph-curves-for-strong-acid-base-neutralisation-reactions-id-46057/>), the use of indicators such as phenolphthalein to indicate the equivalence point was introduced. Did you know that many acid–base indicators can be found in nature or in household materials? Natural indicators are in foods like red cabbage and berries, or in flowers. The varying acidity of the soil can change the colour of some flowers from blue or pink to purple since the flowers contain some type of substance that changes colour with pH. Beautiful displays can be seen in garden in which the soil acidity is changed. Can we make use of such changes to understand more about acid–base reactions?



Figure 1. The different colours of the same flower.

Credit: Amana Images Inc, Getty Images

Watch **Video 1** to find out more about natural indicators.

How to make a home pH indicator from blueberries (si...

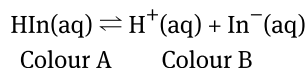


Video 1. How to use a natural indicator to indicate pH.

🕒 More information for video 1

How acid–base indicators work

An acid–base indicator is a weak acid or a weak base in which the dissociated and undissociated forms have different colours. Like all weak acids and weak bases, they exist in an equilibrium between the two forms, which are represented as HIn (the undissociated form) and In^- (the dissociated form) as seen in section R3.1.9–10 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ion-dissociation-constants-of-weak-acids-and-bases-hl-id-46058/>). The equation for the dissociation of HIn can be represented as:



where HIn and In^- have different colours (colour A and colour B). As this is an equilibrium reaction, the position of equilibrium can be changed by adding an acid (H^+ ions) or a base (OH^- ions) [section R2.3.4](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/le-chateliers-principle-id-46069/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/le-chateliers-principle-id-46069/>). Depending on the concentration of H^+ ions in solution (or OH^- ions), the solution takes one colour or the other.

Study skills

The generalised formula HIn(aq) can be used to represent the undissociated form of an indicator.

Figure 2 shows the undissociated and dissociated forms of phenolphthalein, which are colourless and pink respectively.

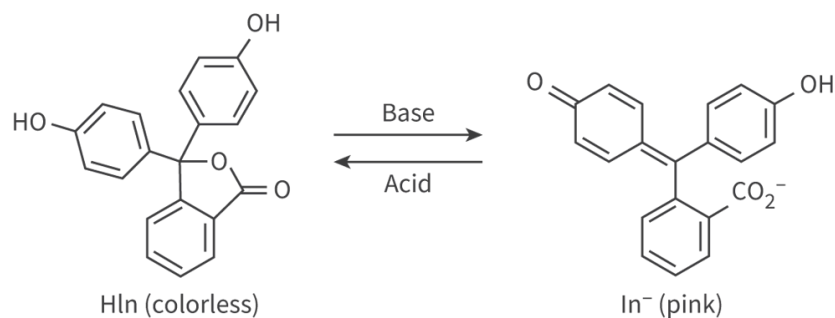


Figure 2. Phenolphthalein in its undissociated (HIn) and dissociated (In^-) forms.

 More information for figure 2

The image depicts the chemical equilibrium between the undissociated (HIn) and dissociated (In^-) forms of phenolphthalein. On the left, the structure of the undissociated form, HIn, is shown as a large molecule with three benzene rings connected by oxygen and hydroxyl groups. This form is noted to be colorless. On the right, the dissociated form, In^- , is depicted as a similar molecular structure but includes an additional carboxylate group (CO_2^-), which breaks the symmetry. This form is indicated to be pink. Arrows labeled 'Base' and 'Acid' indicate the reversible reaction between the two forms based on pH changes.

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Applying Le Chatelier's principle [section R2.3.4](#)

(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/le-chateliers-principle-id-46069/>), we can predict how this equilibrium will respond to a change in pH.

- Increasing $[\text{H}^+]$ by adding an acid will shift the equilibrium to the left, so phenolphthalein appears colourless.
- Increasing $[\text{OH}^-]$ by adding a base will shift the equilibrium to the right, so phenolphthalein appears pink.

To summarise, the addition of an acid shifts the position of equilibrium to the side of the undissociated form (HIn), while the addition of a base shifts the equilibrium to the side of the dissociated form (In^-).

Indicators change colour when their $\text{pH} = \text{p}K_a$.

At equilibrium:

$$K_a = \frac{[\text{H}^+][\text{In}^-]}{[\text{HIn}]}$$

Indicators change colour when $[\text{In}^-] = [\text{HIn}]$ and therefore $K_a = [\text{H}^+]$.

Table 1 shows a list of indicators, their pH ranges for the colour change and the corresponding $\text{p}K_a$ values.

Table 1. A list of indicators, with their pH ranges and $\text{p}K_a$ values.

Indicator	pK_a	pH range	Colour in acid	Colour in
Methyl orange	3.7	3.1–4.4	red	yellow
Bromophenol blue	4.2	3.0–4.6	yellow	blue
Bromocresol green	4.7	3.8–5.4	yellow	blue
Methyl red	5.1	4.4–6.2	red	yellow
Bromothymol blue	7.0	6.0–7.6	yellow	blue
Phenol red	7.9	6.8–8.4	yellow	red
Phenolphthalein	9.6	8.3–10.0	colourless	pink

Study skills

Some acid–base indicators with pH ranges and pK_a values are listed in [section 18 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/acid-base-indicators-id-45120/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/acid-base-indicators-id-45120/) of the DP chemistry data booklet.

Note that indicators do not necessarily change colour around pH 7. Most indicators are weak acids with their own value of pK_a – they change colour over a pH range that centres on that value. For example, the change in colour of bromothymol blue, known as its end-point, happens in the range pH 6.0–7.6, which is within the approximate range of the $pK_a \pm 1.0$.

Study skills

Take care not to confuse the equivalence point and the end-point.

- The end-point is the pH when the indicator changes colour.
- The equivalence point is when acid and base have reacted in stoichiometrically equivalent amounts.

A suitable indicator for a titration is one for which these two points coincide, so the indicator changes colour near to the equivalence point. The indicator is also undergoing its own titration simultaneously to the titration that is the focus of the experiment. The indicator reaches its buffer region for the pH range and when the pH is equal the pK_a , both colours of the indicator are seen – that is the ideal choice for the titration because the colour transition is the one that should show.

Figure 3 shows how the colour change of bromothymol blue is capable of detecting the equivalence point in the titration of a strong alkali with a strong acid. The yellow and blue are both shown in a very brief window, appearing green – the green colour is the colour that should show for the titration – that is where the pH at equivalence

is matching the pK_a of the indicator. The key factor here is that the colour change takes place over a small addition of acid, coinciding with the vertical inflection in the pH curve.

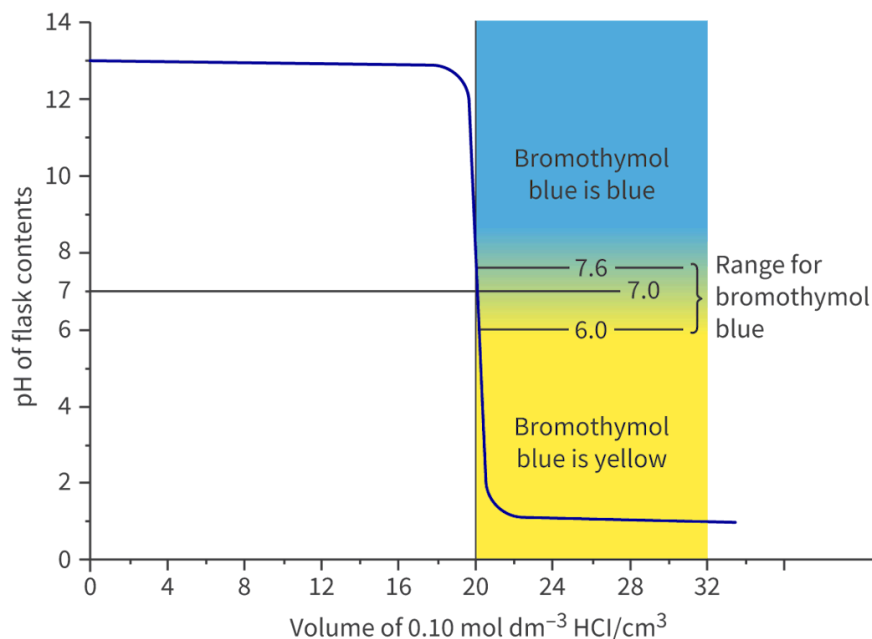


Figure 3. A strong alkali–strong acid titration curve showing the colour change for bromothymol blue.

 More information for figure 3

The graph illustrates a titration curve of a strong alkali reacting with a strong acid, using bromothymol blue as an indicator. The x-axis represents the volume of 0.10 mol/dm^3 HCl (in cm^3), ranging from 0 to 32, while the y-axis represents the pH of the flask contents, ranging from 0 to 14. The curve remains nearly horizontal with a high pH on the left side before sharply dropping near the equivalence point. The vertical drop signifies a rapid pH change due to the addition of acid. The area where the indicator changes color is highlighted: bromothymol blue is yellow below pH 6.0 and blue above pH 7.6, with the equivalence range marking a transition around pH 7.0. The pH change over a small volume of acid addition is critical, aligning with the steep inflection of the curve.

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Figure 4 shows the relationship between the pH range over which phenolphthalein changes colour and the pH curve for a strong alkali–strong acid titration. This shows that even though phenolphthalein changes colour at approximately pH 9.6, it is still an appropriate indicator for this titration. This is because the range over which the colour changes falls within the vertical region of the curve.

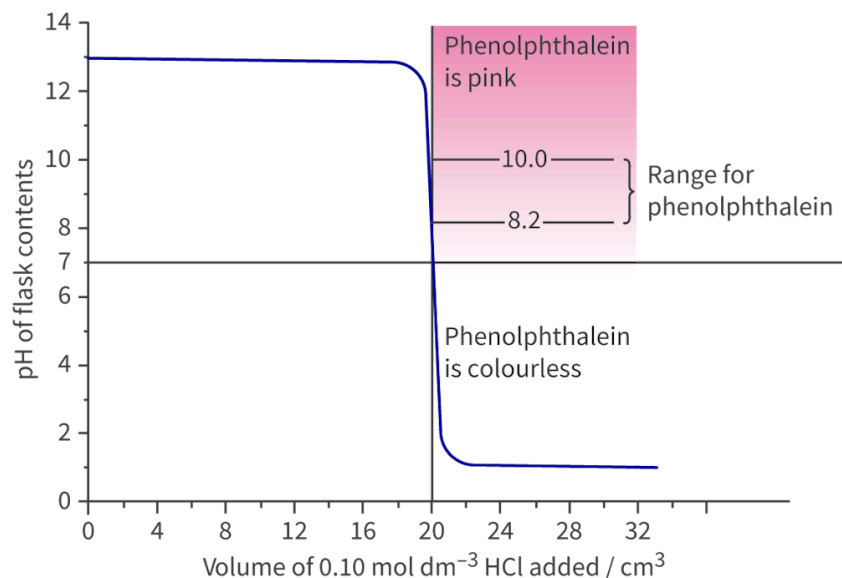


Figure 4. A strong alkali—strong acid titration curve using phenolphthalein.

 More information for figure 4

The graph shows a titration curve for a strong alkali-strong acid with phenolphthalein as an indicator. The X-axis represents the volume of 0.10 mol dm⁻³ HCl added in cm³, ranging from 0 to 34. The Y-axis represents the pH of the flask contents, ranging from 0 to 14.

The curve starts at a high pH level around 14 and remains level for the initial part until approximately 20 cm³ of HCl is added. At this point, the curve sharply declines, indicating the equivalence point and a significant pH change within a small volume, common in strong acid-strong base reactions.

Phenolphthalein's color change range, between 8.2 and 10.0 pH, is marked on the graph. In this range, phenolphthalein appears pink. Below 8.2 pH, it is colorless, indicating the endpoint of the titration.

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Figure 3 and **Figure 4** show that both phenolphthalein and bromothymol blue would be appropriate indicators for titrations between a strong acid and a strong alkali. In fact, the size of the pH change around the equivalence point in such titrations is sufficiently great that several different indicators could be used.

Looking at the titration curves for other acid–base combinations, it is possible to make a judgement as to which indicators are suitable to use for a given titration.

Figure 5 shows that while phenolphthalein is an appropriate indicator for a titration of a strong base with a weak acid, methyl orange is not. This is because the pH range over which it changes colour (its end-point) does not fall within the vertical region of the curve.

Remember that at the equivalence point, the acid and base have reacted completely, and the salt has been formed. The pH at the equivalence point is the pH of the salt. Therefore, to find a suitable indicator for a titration:

1. Determine which acid and base are reacting.
2. Find the pH of the salt at the equivalence point.

3. Choose an indicator with an end point in the range of the equivalence point or close enough to it, i.e., where its pK_a is equal or close enough to the pH at the equivalence point.

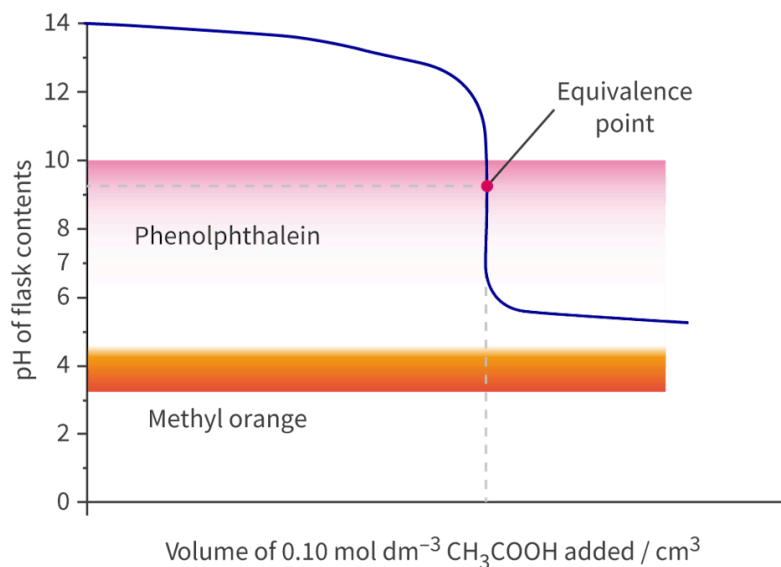


Figure 5. A strong alkali—weak acid titration curve using phenolphthalein and methyl orange.

 More information for figure 5

The image displays a titration curve with a blue line representing the change in pH as a strong alkali (base) is titrated with a weak acid. The X-axis shows the volume of 0.10 mol dm⁻³ CH₃COOH added in cm³, while the Y-axis shows the pH of the flask contents ranging from 0 to 14. The curve exhibits a steep rise near the equivalence point.

The graph highlights two indicators: "Phenolphthalein" and "Methyl orange," each marked by shaded areas on the pH scale. "Phenolphthalein" ranges from around pH 8 to pH 10, showing where it is effective as a pH indicator, while "Methyl orange" covers pH 3 to pH 4.

The "Equivalence point" is labeled on the curve, marked by a red dot where the slope is steepest, indicating the point at which stoichiometrically equivalent amounts of acid and base have reacted. This point is important for choosing the correct indicator to determine the transition colour close to the equivalence point, as noted in the additional context about detecting pale pink with phenolphthalein.

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Remember that you want the indicator to go through the colour change near the equivalence point. You want to see the transition colour at the equivalence point. For phenolphthalein, you will be able to detect the equivalence point if the end point or the transition colour (pale pink) coincides with the equivalence point.

Use **Interactive 1** to match different acid/base titrations with a suitable indicator.

Using the information on indicators in the data booklet:

What is a suitable indicator for a strong acid–strong base titration?

 Turn

Card 1 of 4



Interactive 1. Review choice of suitable indicator.

Making connections

What are some of the similarities and differences between indicators used in acid–base titrations and in redox titrations? (see [section R3.2](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43486/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43486/>)).

Activity

- **IB learner profile attribute:** Thinker
- **Approaches to learning:**
 - Thinking skills — Being curious about the natural world; Applying key ideas and facts in new contexts
 - Social skills —Working collaboratively to achieve a common goal
- **Time required to complete activity:** 20 minutes
- **Activity type:** Pair activity

Instructions

Go on a hunt!

1. Find at least four chemical indicators at home or used for industrial purposes.
2. Include their names and their uses.
3. Find their pK_a and pH range values and include the equilibrium equation using correct formulas of the acid and ionised form of the indicator.

Conclude with justification for which titration (strong acid—strong base, strong acid—weak base or weak acid—strong base) they could be used in order to indicate the equivalence point.